

PTTEP: Accelerating sustainable energy innovations with application modernization

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ABOUT PTTEP



PTTEP has executed complete application modernization in 2021, created an iAPI Hub, and achieved seamless database compatibility across business functions using Apigee solutions.

About PTTEP

PTT Exploration and Production Public Company Limited or PTTEP is Thailand's national oil and gas company engaged in petroleum exploration, development, and production. Founded in 1985, PTTEP plays a crucial role in supporting Thailand's energy security and the energy demand of countries we operate in, while creating value for all stakeholders. Currently, PTTEP invests in over 50 petroleum exploration, development, and production projects worldwide, covering Southeast Asia, the Middle East, Africa, and Americas. In preparation for

Google Cloud results

- Drives energy innovation strategy by reducing API development time from at least three days to five hours
- Accelerates operational processes with instant one-click API deployment
- Enhances scalability and agility through API reusability
- Ensures system and business continuity by fixing API issues within 2 hours

Comp applic mode under

energy transition, PTTEP has diversified into new businesses involving new forms of energy, advanced technology and decarbonization while emphasizing on carbon emissions reduction as part of EP Net Zero 2050 pathway.

Industries: Oil & Gas

Location: Thailand

Functions

Google Cloud (<https://cloud.google.com/>)

Kubernetes Engine

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Apigee API Monitoring

(<https://cloud.google.com/apigee/api-management/monitor-apis/>)

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(<https://www.pttep.com/en/Home.aspx>) is Thailand's national oil and gas company engaged in petroleum exploration, development, and production. PTTEP plays a crucial role in supporting Thailand's energy security and the energy demand of countries they operate in, while creating value for all stakeholders. In preparation for energy transition, PTTEP has diversified into new businesses involving new forms of energy, advanced technology and decarbonization while emphasizing on carbon emissions reduction as part of EP Net Zero 2050 pathway.

Through machine learning (ML) modeling, PTTEP can analyze, predict, and optimize energy exploration that spans the globe. It is a complex enterprise that requires data processing and integration from five principal business functions: data asset production, drilling and well services, engineering and maintenance, subsurface operations, and support functions, including human resources, legal, and information technology.



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*—Sagita Suwandi, Head
of Software Development,
PTTEP*

In this sprawling organization, data traditionally remained siloed between and within business functions themselves. Integrations were typically done as direct application to application, and often were not compatible. Some were so old they did not have APIs. Driving a unified big data-powered strategy, encompassing all five PTTEP functions, is thus an overwhelming task.

These issues inspired PTTEP to embark upon a comprehensive API-first application modernization program. The goal was to build an API Hub as a one-stop-shop for all APIs so that authorized parties within the organization could securely exchange data in real-time, to drive unlimited scale for ML-powered exploration analytics and solution building.

PTTEP chose Apigee (<https://cloud.google.com/apigee>) as the central foundation for its application modernization project, thanks to the Google Cloud (<https://cloud.google.com/>) API management platform's intuitive interface and what it viewed as best-in-class API gateway, lifecycle management, developer portal, and security features. The result was a fully modernized API-first application architecture within one year, enabling PTTEP to accelerate its mission of sustainable energy business value.

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Driving scalability and innovation in energy exploration through API modernization

Before adopting Apigee, PTTEP attempted to achieve database integration using APIs developed for older applications on a laborious case-by-case basis. For more modern applications, it deployed an open-source platform as its API gateway to act as a go-between between API clients and backend services. When embarking on a full-fledged app modernization program, PTTEP decided it needed a more powerful and versatile API management solution, leading to its choice of Apigee.

Legacy applications were rapidly replaced by new ones built on a Google Cloud architecture based on Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE)

microservices, with Cloud Functions

(<https://cloud.google.com/functions>) as its serverless compute platform, and BigQuery

(<https://cloud.google.com/bigquery>) for ML-driven data analytics. Apigee served as the glue that enabled these new modernized apps to work together in seamless, coordinated fashion.



PTTEP employee working on a complex issue

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—*Jaru Sahassakunakorn,*
Software Developer,
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According to Suwandi, one of the big advantages PTTEP gained from Apigee has been API reusability. The ability to reuse APIs reduces development time by allowing engineers to build new apps from existing code, instead of starting from scratch. It also helps businesses scale development by allowing them to create on top of APIs, expanding their functionality according to priorities. This unlocked the agility and solution scalability the company was seeking.

"Apigee has transformed the speed of our innovation and ability to make continuous app enhancements, through API reusability that enables an optimized developer experience," says Jaru Sahassakunakorn, Software Developer, PTTEP. "And with Apigee, everything is automated so we can focus on the business with no need to worry about setups and other mundane tasks."

With Apigee, PTTEP has been able to create a vast library of APIs for its API Hub quickly, which is now serving as a single repository for all authorized members of the corporate team to share information in a seamless and collaborative manner. PTTEP also uses another cloud provider for different aspects of its operational needs. Apigee enables the frictionless integration of apps in this hybrid environment, through APIs that serve as a go-between for the two public clouds.

The results of PTTEP's API-led transformation have been impressive. With Apigee, PTTEP has reduced the time needed for API development from code to development and production from at least three days to five hours, for a 480% improvement in speed. Launching APIs is now achieved instantly with one click in Apigee, whereas in the past deployment was a multi-step process. And with Apigee's sophisticated lifecycle management features, PTTEP can fix issues arising with APIs within two hours, something that normally required a week in the past.

Enabling dynamic data-sharing through agile and resilient security protocols

For an energy exploration company with proprietary technologies in a competitive industry, security is of the utmost importance. Apigee offers multiple security layers for granular access control on APIs that are easy to set up and disable, says Suwandi. PTTEP also likes the way Apigee automates the securing of backend applications so engineers are

freed to focus on solution building. Suwandi estimates that Apigee automation saves the firm 15 hours per week in DevOps time.

PTTEP is further bolstering the security of its API infrastructure through a feature of Apigee that enables the company to set API key expirations to protect it from hacking attacks. The company has set key expiration to once a month giving it the peace of mind to continue to freely operate its API Hub and develop and launch new innovative applications.

"With the elastic
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energy solutions
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—Wuttichai Srisuk, Senior
Software Developer,
PTTEP

Looking to the future, PTTEP plans to enhance its AI and ML modeling capabilities for the prediction and analysis of oil and gas exploration opportunities through adopting Vertex AI (<https://cloud.google.com/vertex-ai>). This promises to open up new, secure, and dynamic paths to innovation for a firm striving to be at the forefront of sustainable energy supply for Thailand and beyond.

"With the elastic and agile API development and secure data exchange possibilities delivered by Apigee, alongside exciting new AI projects we foresee with Vertex AI, we envision an unlimited future for us to deliver sustainable energy solutions for Thailand's economic development and wellbeing," says Wuttichai Srisuk, Senior Software Developer, PTTEP.

